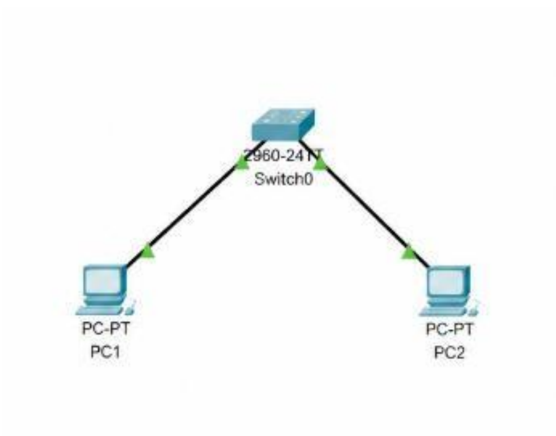
Switching Basics Theory: Switch operation: MAC address table, frame forwarding. Practical: Observe MAC table population in a simulated network. Task: Explain the process of frame forwarding in switches.

Q-1 Open Cisco Packet Tracer or any network simulator, create a simple network with one switch and two PCs, and use the 'show mac address-table' command to observe how the MAC address table is initially empty.

[A]




```
Switch>enable
Switch#show mac address-table

                    Mac Address Table
             -------------------------------

Vlan      Mac Address          Type        Ports
----      -----------------    --------    -----

Total Mac Addresses for this criterion: 0

Switch#
```

Network Created:

- 1 Switch
- PC1
- PC2
- PC1 connected to Switch Port 1
- PC2 connected to Switch Port 2

Initial MAC Address Table

- When the switch is newly connected and no communication has happened yet, the MAC address table is initially empty or contains very few learned entries. - Command used on the switch : show mac address-table

- **Observation:**
- Initially, the switch does not know which MAC address belongs to which port. It learns MAC addresses only when it receives Ethernet frames from the connected devices.

Q-2 Send a ping from PC1 to PC2 in your simulated network, then immediately run 'show mac address-table' on the switch and note how the MAC addresses of both PCs appear in the table. Hint: Try pinging in both directions to see how the table updates

[A]

First, assign IP addresses: PC1:
192.168.1.1
PC2: 192.168.1.2
Subnet Mask: 255.255.255.0 From
PC1, run:
ping 192.168.1.2
After the ping, check the switch:
show mac address-table

```
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Observation:

The switch now learns the MAC addresses of the PCs.

Q-3 Add a third PC to your network, connect it to the switch, and ping from PC3 to PC1. Observe and record how the MAC address table changes after this communication.

[A]

Now add PC3 to the switch. Example:

PC1 → Switch Port 1

PC2 → Switch Port 2

PC3 → Switch Port 3 Assign

PC3:

PC3: 192.168.1.3

Subnet Mask: 255.255.255.0 From

PC3, ping PC1:

ping 192.168.1.1

Then check the MAC table:

show mac address-table

```
Switch>enable
Switch#show mac address-table

                        Mac Address Table
                     ----------------------

Vlan      Mac Address             Type          Ports
----      -------------------     --------      -----
 1        00:11:22:33:44:55       DYNAMIC       Fa0/1
 1        00:AA:BB:CC:DD:EE       DYNAMIC       Fa0/2
 1        00:CC:CC:CC:CC:CC       DYNAMIC       Fa0/3

Total Mac Addresses for this criterion: 3
```

Observation:

The switch will learn PC3's MAC address on Port 3.

Q-4 Suppose you see the following MAC address table on a switch: Port 1 - 00:11:22:33:44:55, Port 2 - 66:77:88:99:AA:BB. If a frame arrives at the switch on Port 1 with destination MAC 66:77:88:99:AA:BB, explain step-by-step how the switch will forward this frame.

[A]

Given MAC address table:

Port 1 → 00:11:22:33:44:55

Port 2 → 66:77:88:99:AA:BB

A frame arrives at **Port 1** with destination MAC : 66:77:88:99:AA:BB

Step-by-Step Process:

Step 1: The switch receives the frame on **Port 1**.

Step 2: The switch checks the **source MAC address** of the frame and can learn/update which port that source MAC is connected to.

Step 3: The switch looks at the **destination MAC address** : 66:77:88:99:AA:BB

Step 4: The switch searches its MAC address table and finds : 66:77:88:99:AA:BB → Port 2

Step 5: Because the destination is known, the switch forwards the frame **only through Port 2**.

Step 6: The device connected to Port 2 receives the frame and processes it.

Q-5 Draw a diagram showing how a switch learns and uses MAC addresses to forward frames in a network with 4 devices, and write a short explanation (3-4 sentences) describing the process in your own words. Constraint : Do not copy definitions from the textbook; use your own understanding and examples.

[A]

```
           PC1
     MAC: AA:AA:AA:AA:AA:01
             |
             | Port 1
             |
             ▼
  +--------------------+
  |    SWITCH          |
  |                    |
  |   MAC TABLE        |
  +--------------------+
      |      |      |
   Port 2  Port 3  Port 4
 |     |      |
      ▼      ▼      ▼
     PC2    PC3    PC4
     MAC:   MAC:   MAC:
    AA:02  AA:03  AA:04
```